

SHRESTH JAISWAL

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

EDUCATION

B.Tech in Computer Science and Engineering — Vellore Institute of Technology

Sep 2023 – Present

CGPA: 8.40 / 10.0

SKILLS

AI/ML & LLMs: LLM APIs (OpenAI GPT-4o, Gemini), Prompt Engineering, Conversational AI, Agentic Workflows (Function Calling, Tool Use), AI Output Evaluation

Voice AI: Pipecat, Deepgram Nova-2 (STT), ElevenLabs (TTS), Twilio Voice, WebRTC, Silero VAD

Languages: Java, Python, JavaScript, SQL

Backend: Django, FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Alembic

Databases: PostgreSQL, SQL

Frontend: React.js, Redux Toolkit, React Hook Form, HTML, CSS, TailwindCSS

Testing & Automation: Pytest, Selenium, API Testing, Unit Testing, Integration Testing

DevOps & Tools: Git, GitHub Actions, Docker, Linux

EXPERIENCE

Software Engineering Intern — Scramble Labs Technology Pvt. Ltd.

May 2025 – Aug 2025

- Engineered 2 production-ready REST APIs using Python, Django, and PostgreSQL, integrating request validation, database optimization, and RESTful architecture to support scalable backend services.
- Developed a Git workflow that standardized code merging for the IT department, preventing 15 critical merge conflicts and improving team collaboration.
- Integrated a comprehensive suite of 15 automated unit and integration tests for newly developed REST APIs.
- Developed 4 modular React.js + TypeScript components within a production Next.js application, reducing page load latency by 15% through React component caching and memoized render hooks.

PROJECTS

EstateStream AI — Real-Time Voice AI Agent for Real-Estate Sales

Python, FastAPI, Pipecat AI, OpenAI GPT-4o, Deepgram Nova-2, ElevenLabs, Twilio, PostgreSQL, Docker

- Built and configured an inbound voice AI agent that qualifies real-estate leads over live phone calls, achieving <800ms end-to-end conversational latency by chaining streaming Deepgram Nova-2 (STT), GPT-4o, and ElevenLabs Turbo v2.5 (TTS) via Pipecat.
- Integrated the agent with Twilio Media Streams for live PSTN broker/customer calls over 8kHz μ -law WebSockets, plus a Web Audio API browser client with real-time waveform visualization.
- Designed conversation flows and guardrails, and implemented native GPT-4o function calling to query live property inventory and CRM data, persisting structured buyer profiles with a 0% hallucination rate.
- Tested the agent across 19 automated scenarios spanning 10 adversarial failure modes (barge-in, long silences, hostile de-escalation) and built an async SQLAlchemy/Alembic pipeline that auto-generates call summaries and outcome tags for the CRM.

TestPilot — AI-Powered Test Automation Platform

Python, FastAPI, Pytest, Selenium, PostgreSQL, LLM API, Docker | [Code](#) | [Live Demo](#)

- Designed structured prompts and engineered an AI-driven test generation pipeline that converts API specifications into executable test cases, covering functional, negative, and edge-case scenarios.
- Built an LLM evaluation framework to assess AI-generated test cases across correctness, consistency, coverage, and practical usability, enabling measurable quality scoring.
- Automated API and UI validation using Pytest and Selenium, with reusable test workflows for authentication, response schemas, status codes, and critical user journeys.
- Containerized and productionized the testing platform with Docker, PostgreSQL, and GitHub Actions for reproducible environments, automated test execution, and centralized test-result tracking.

AssetFlow ERP — Enterprise Asset & Lifecycle Management Platform

PostgreSQL, Express.js, React.js, Node.js, Knex.js, REST API | [Code](#) | [Live Demo](#)

- Architected an enterprise-grade platform for equipment tracking, checkout workflows, and maintenance requests, with 4 granular role-based permission tiers and a PostgreSQL schema spanning 10+ normalized entities.
- Built client-side caching (SWR) with asynchronous parallel API execution in React, and enforced transaction audit logging with payload validation across Express.js/Knex.js controllers.
- Achieved a 3x speedup on complex query execution and sub-second rendering across 9 dashboard sections, with full historical data integrity.

ACHIEVEMENTS & LEADERSHIP

- Data Structures & Algorithms: Solved 240+ Data Structures & Algorithms problems on LeetCode.
- Community Leadership: Core Member of the Notion Community at VIT Bhopal — organized tech talks and peer learning workshops.
- Professional Certifications: Earned NPTEL Silver Elite Certification in Cloud Computing & Internet of Things (IoT).